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Data and Computer

Communications

7th Edition

Chapter 2

Protocols and Architecture

Need For Protocol Architecture

- E.g. File transfer
 - Source must activate comms. Path or inform network of destination
 - Source must check destination is prepared to receive
 - File transfer application on source must check destination file management system will accept and store file for his user
 - May need file format translation
- Task broken into subtasks
- Implemented separately in layers in stack
- Functions needed in both systems
- Peer layers communicate

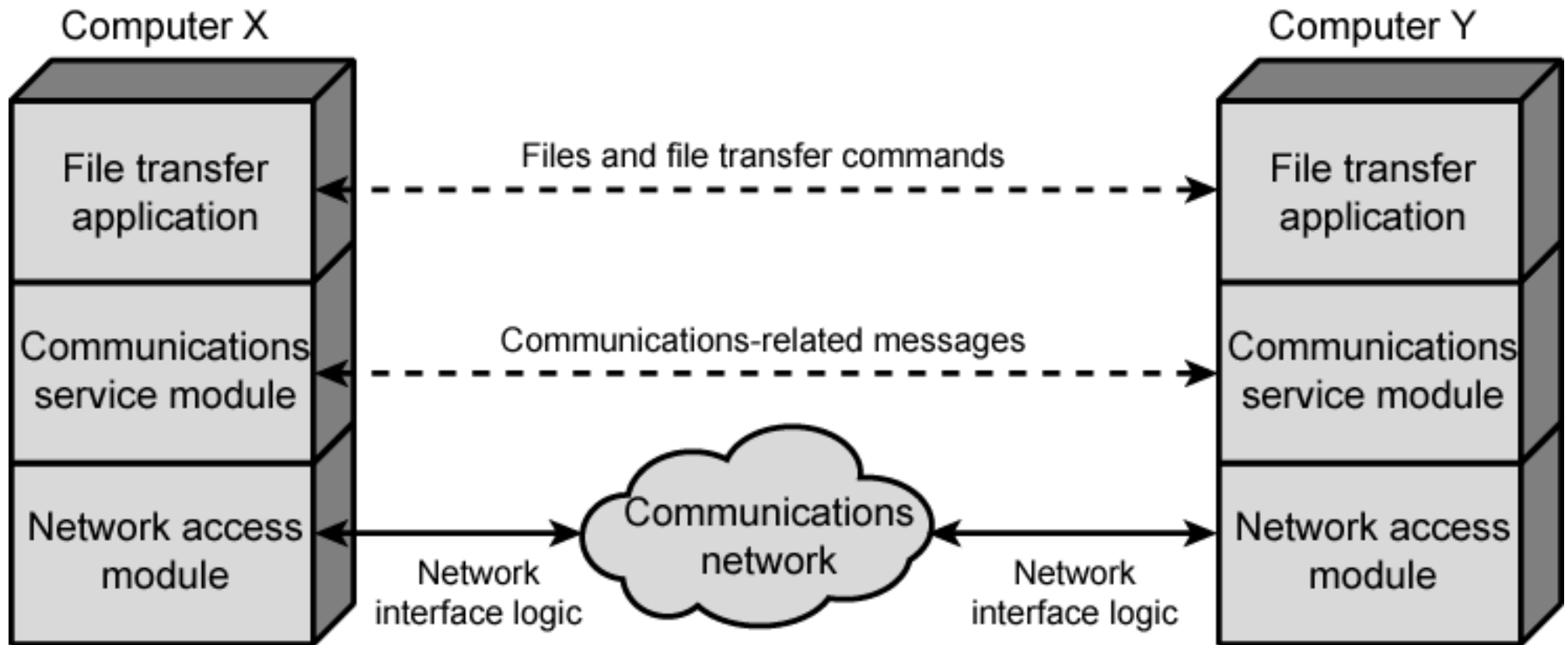
Key Elements of a Protocol

- Syntax
 - Data formats
 - Signal levels
- Semantics
 - Control information
 - Error handling
- Timing
 - Speed matching
 - Sequencing

Protocol Architecture

- Task of communication broken up into modules
- For example file transfer could use three modules
 - File transfer application
 - Communication service module
 - Network access module

Simplified File Transfer Architecture



A Three Layer Model

- Network Access Layer
- Transport Layer
- Application Layer

Network Access Layer

- Exchange of data between the computer and the network
- Sending computer provides address of destination
- May invoke levels of service
- Dependent on type of network used (LAN, packet switched etc.)

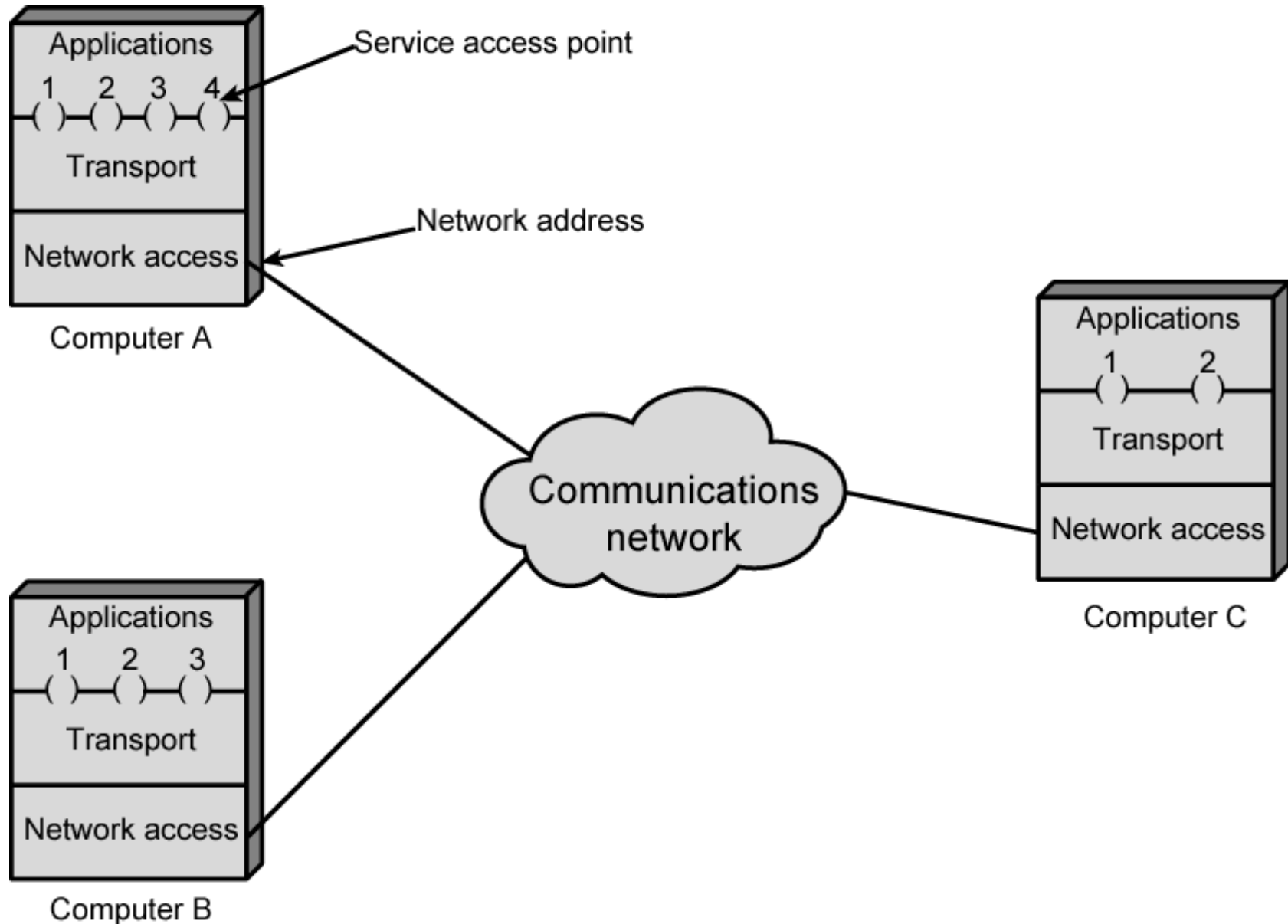
Transport Layer

- Reliable data exchange
- Independent of network being used
- Independent of application

Application Layer

- Support for different user applications
- e.g. e-mail, file transfer

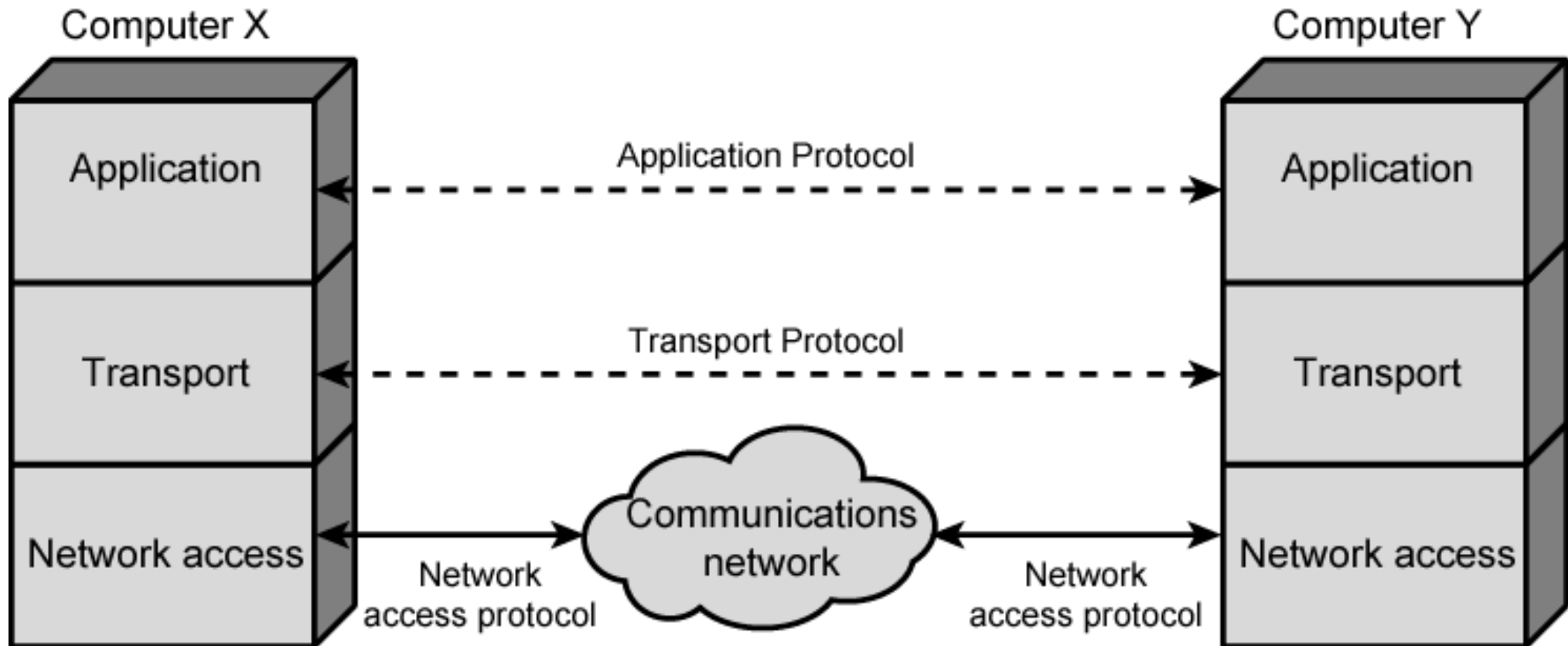
Protocol Architectures and Networks



Addressing Requirements

- Two levels of addressing required
- Each computer needs unique network address
- Each application on a (multi-tasking) computer needs a unique address within the computer
 - The service access point or SAP
 - The port on TCP/IP stacks

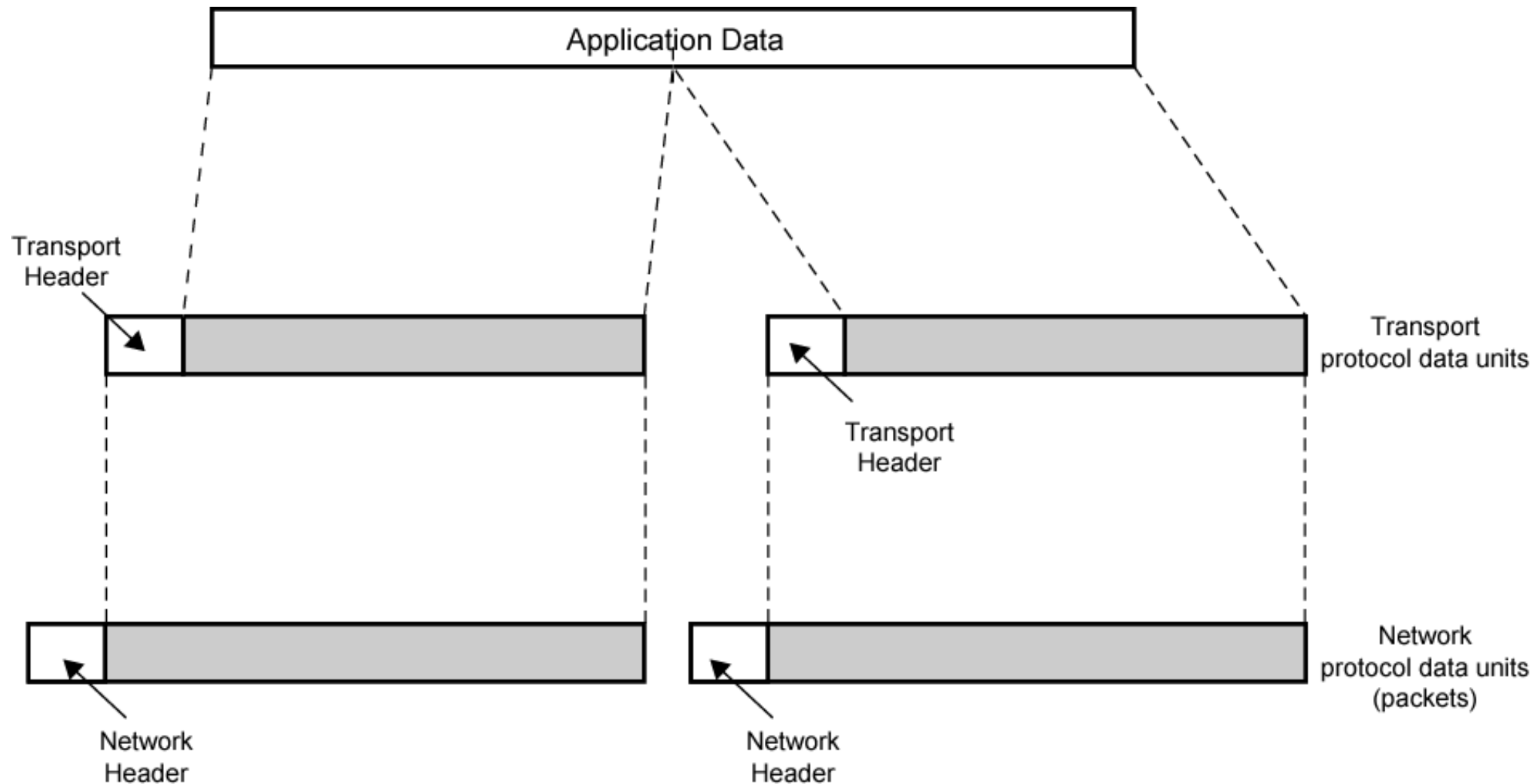
Protocols in Simplified Architecture



Protocol Data Units (PDU)

- At each layer, protocols are used to communicate
- Control information is added to user data at each layer
- Transport layer may fragment user data
- Each fragment has a transport header added
 - Destination SAP
 - Sequence number
 - Error detection code
- This gives a transport protocol data unit

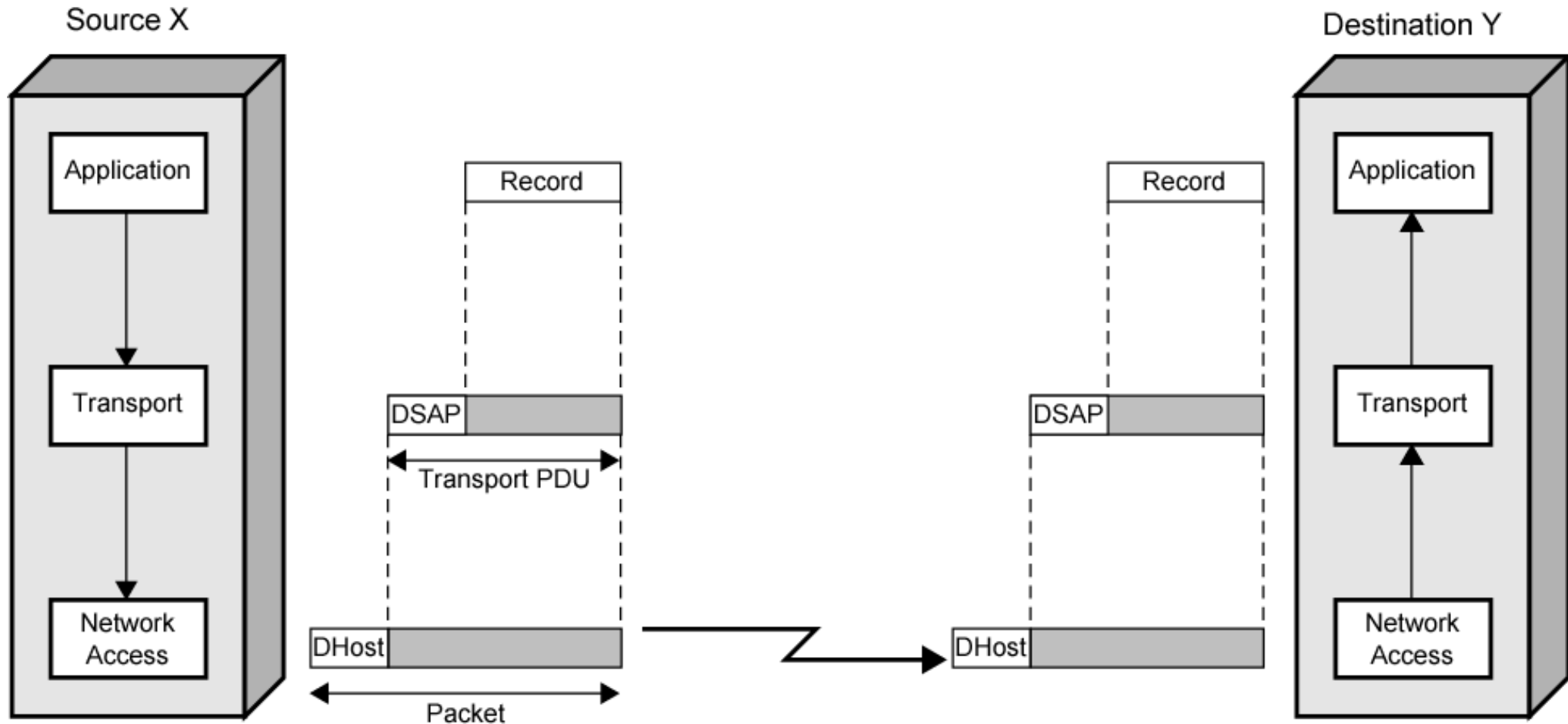
Protocol Data Units



Network PDU

- Adds network header
 - network address for destination computer
 - Facilities requests

Operation of a Protocol Architecture



Standardized Protocol Architectures

- Required for devices to communicate
- Vendors have more marketable products
- Customers can insist on standards based equipment
- Two standards:
 - OSI Reference model
 - Never lived up to early promises
 - TCP/IP protocol suite
 - Most widely used
- Also: IBM Systems Network Architecture (SNA)

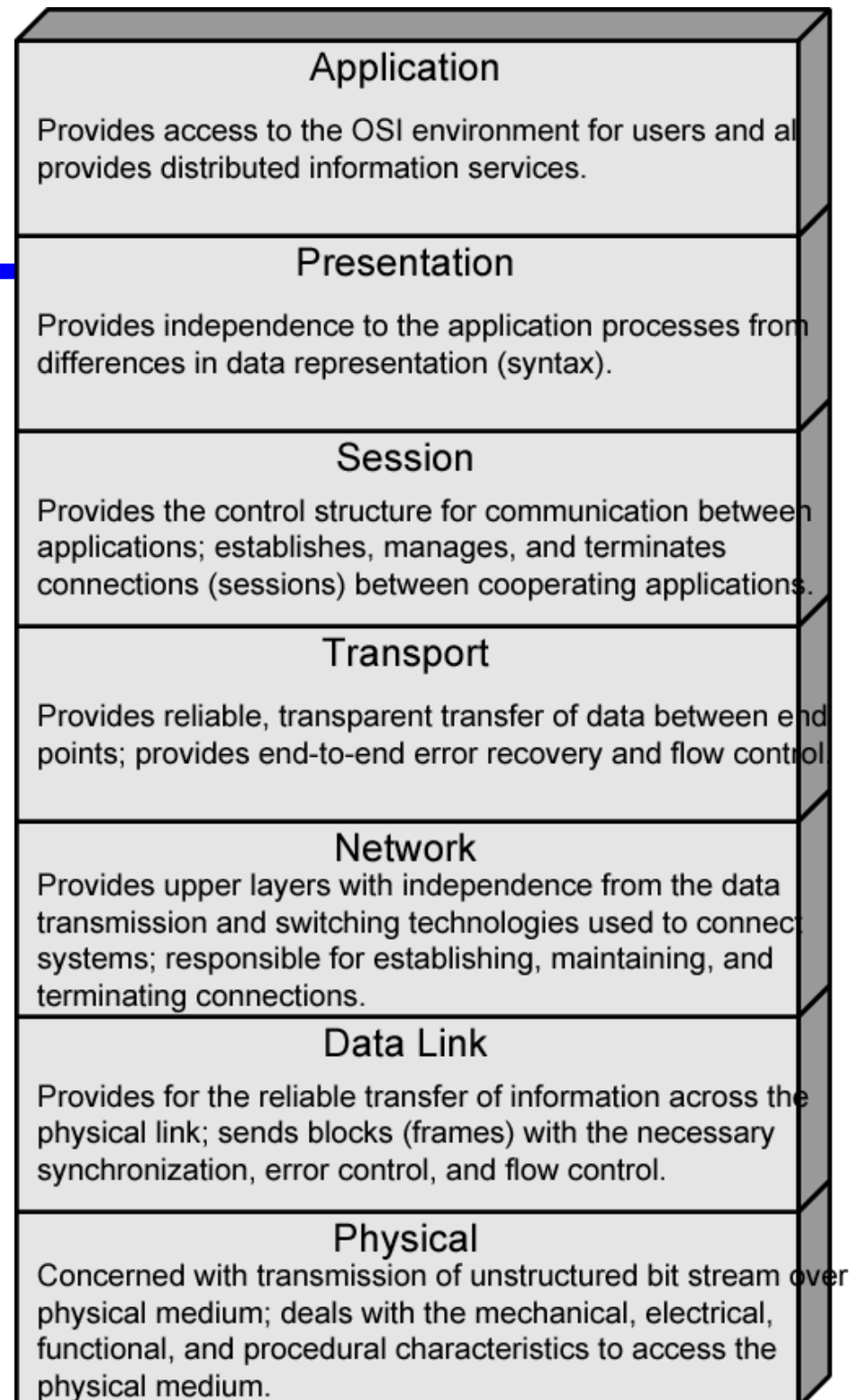
OSI

- Open Systems Interconnection
- Developed by the International Organization for Standardization (ISO)
- Seven layers
- A theoretical system delivered too late!
- TCP/IP is the de facto standard

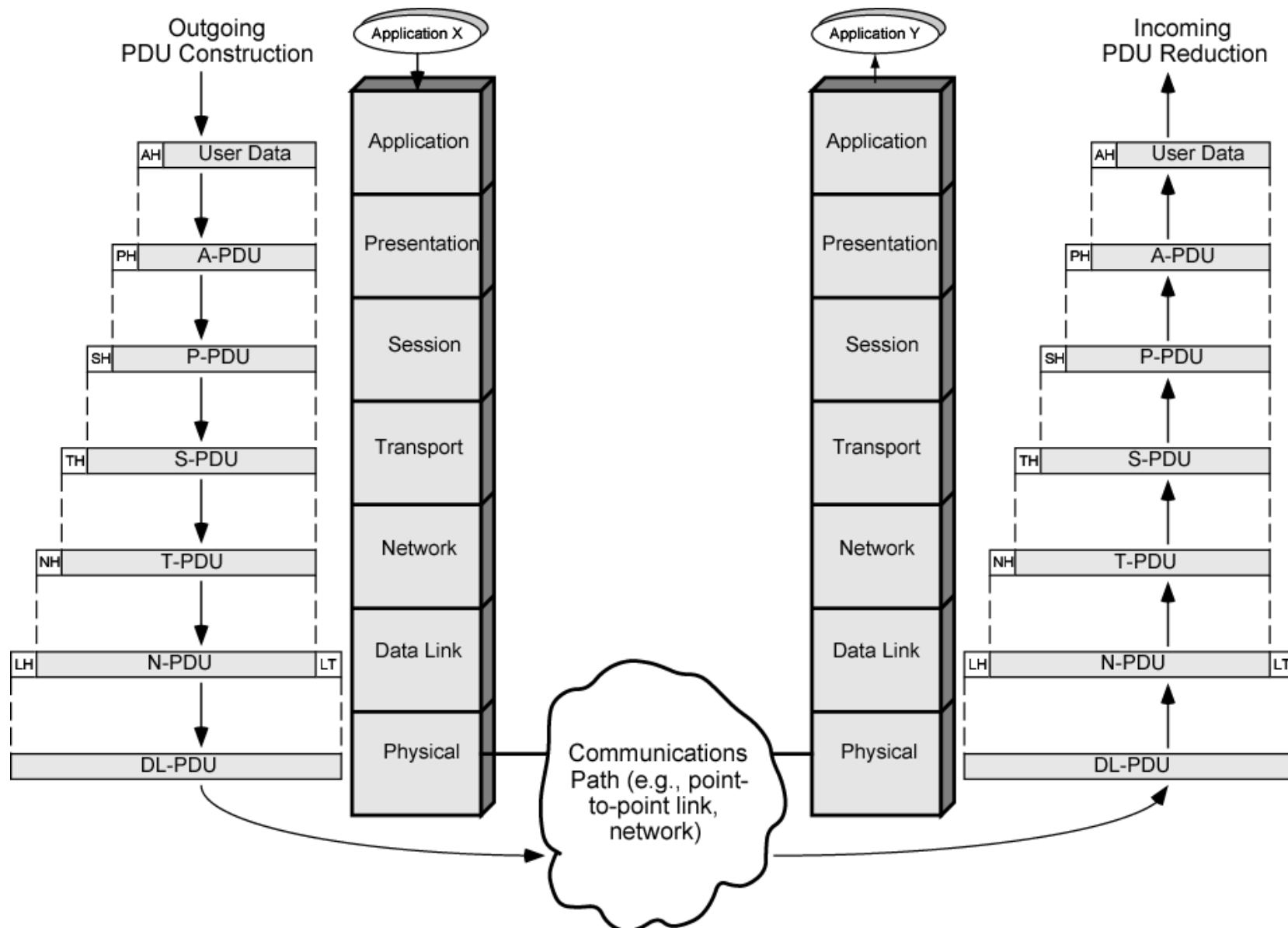
OSI - The Model

- A layer model
- Each layer performs a subset of the required communication functions
- Each layer relies on the next lower layer to perform more primitive functions
- Each layer provides services to the next higher layer
- Changes in one layer should not require changes in other layers

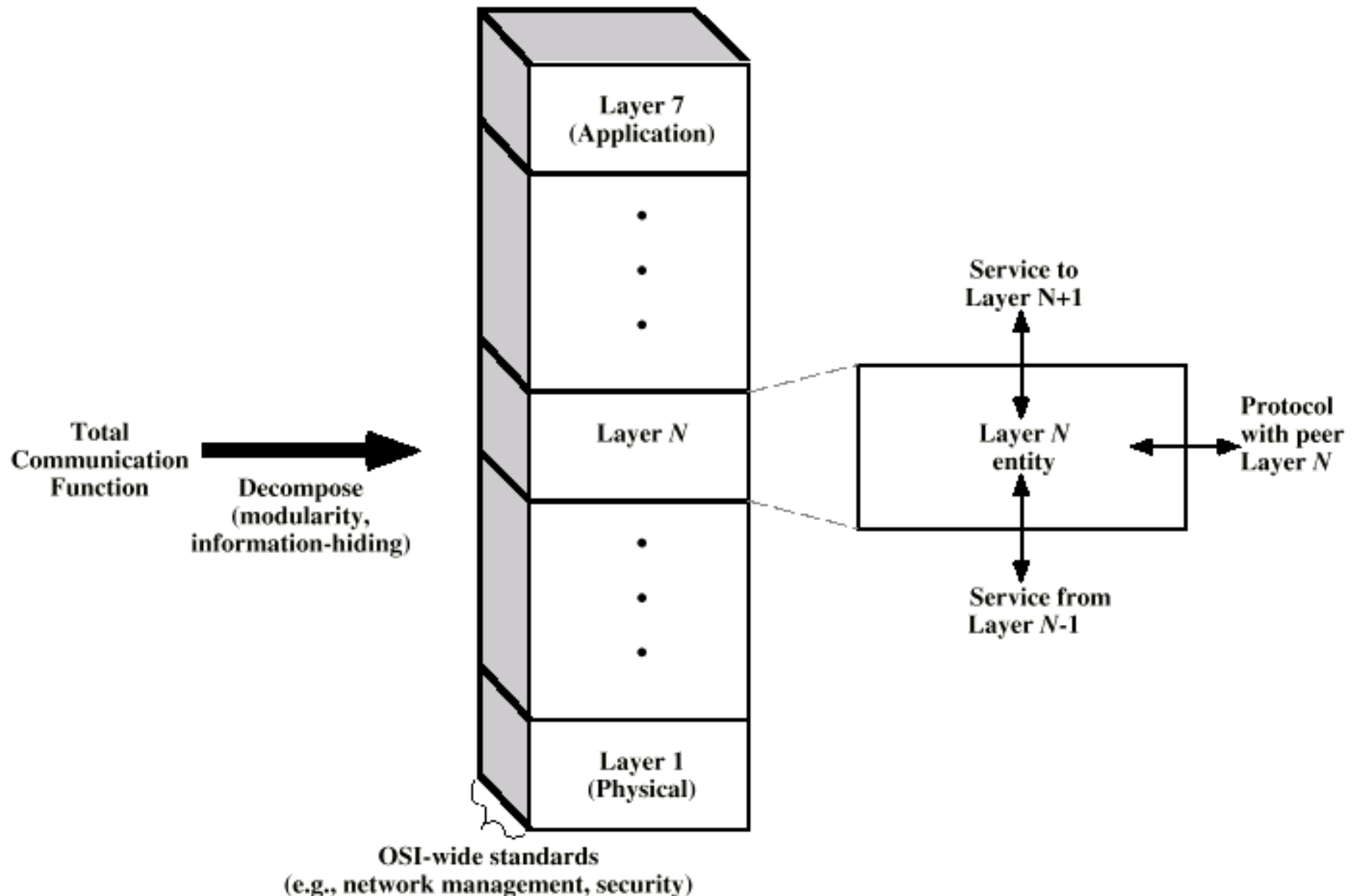
OSI Layers



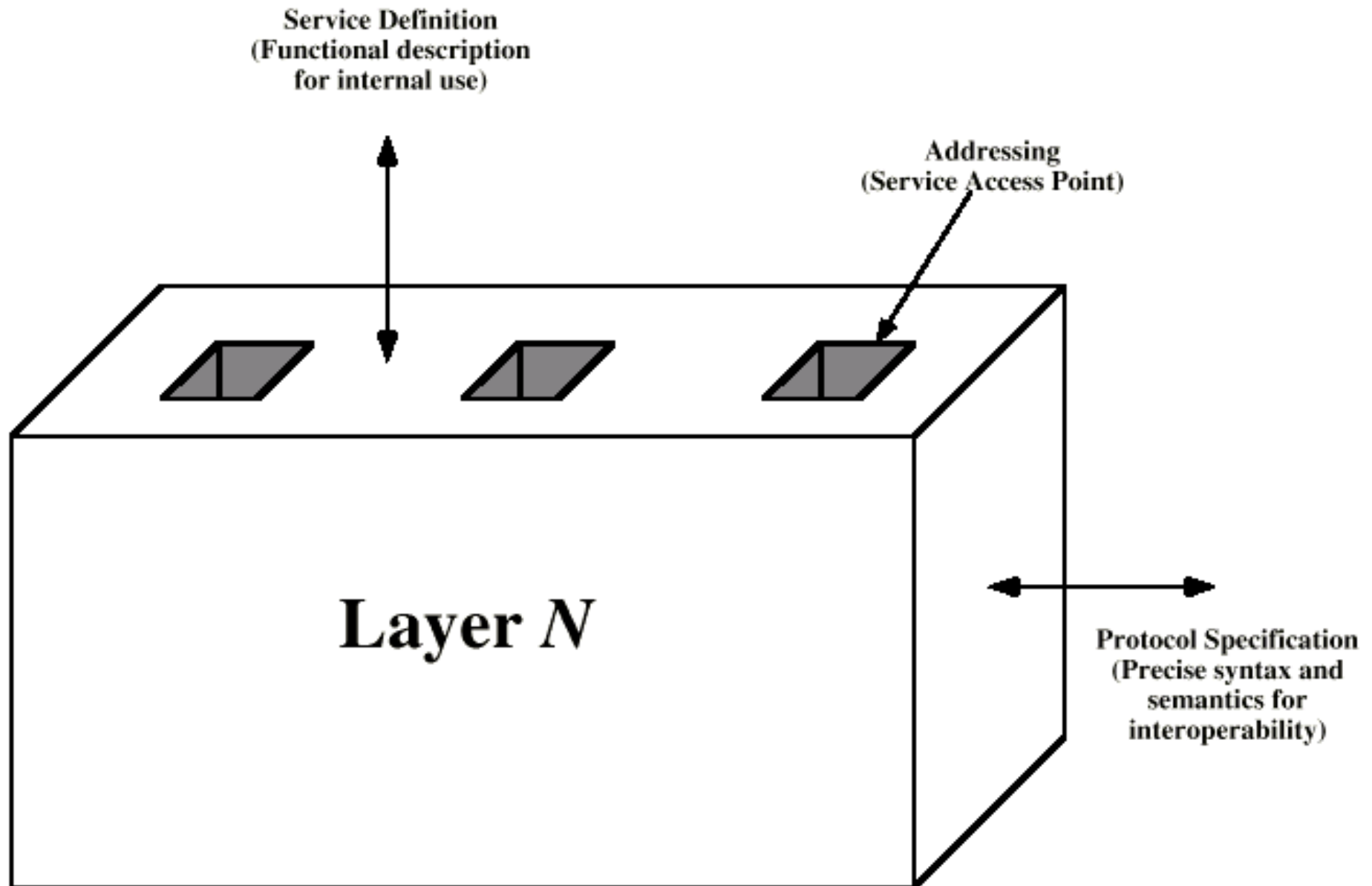
The OSI Environment



OSI as Framework for Standardization



Layer Specific Standards



Elements of Standardization

- Protocol specification
 - Operates between the same layer on two systems
 - May involve different operating system
 - Protocol specification must be precise
 - Format of data units
 - Semantics of all fields
 - allowable sequence of PCUs
- Service definition
 - Functional description of what is provided
- Addressing
 - Referenced by SAPs

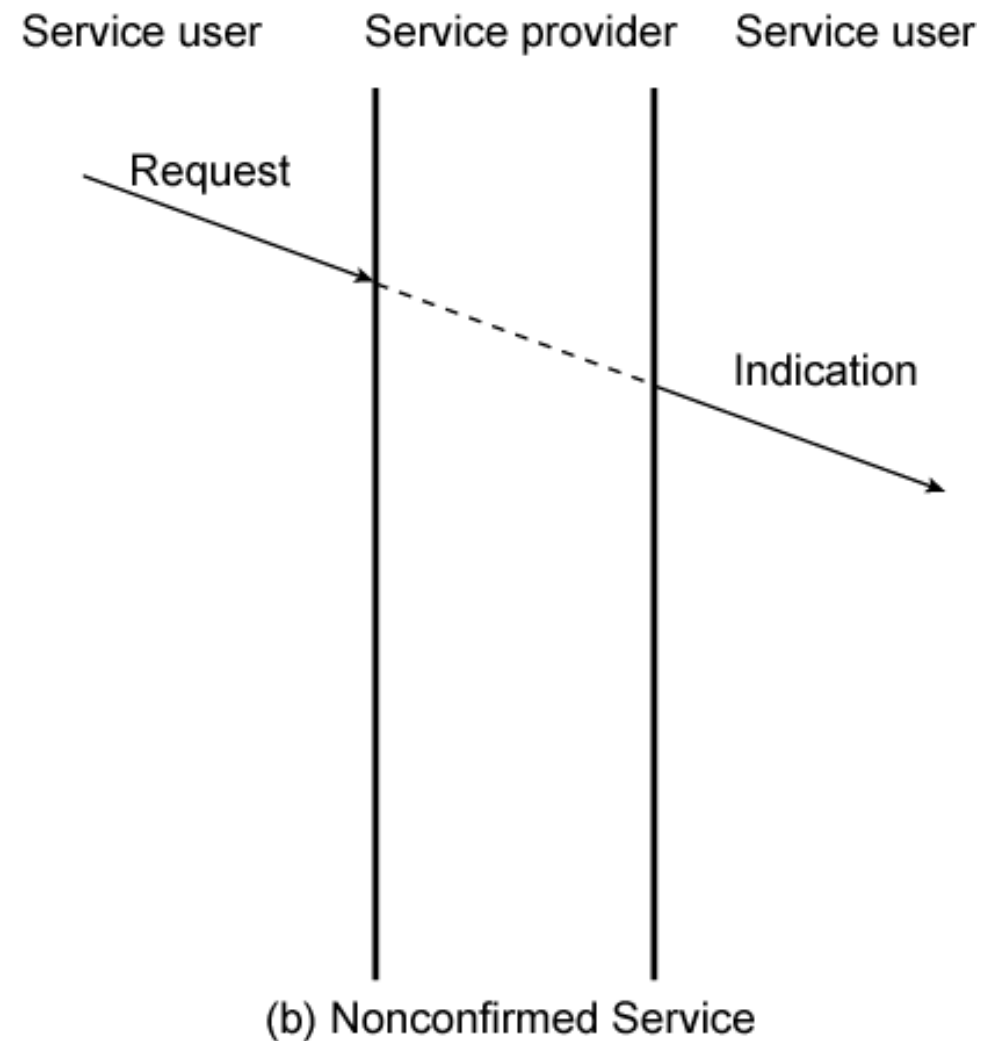
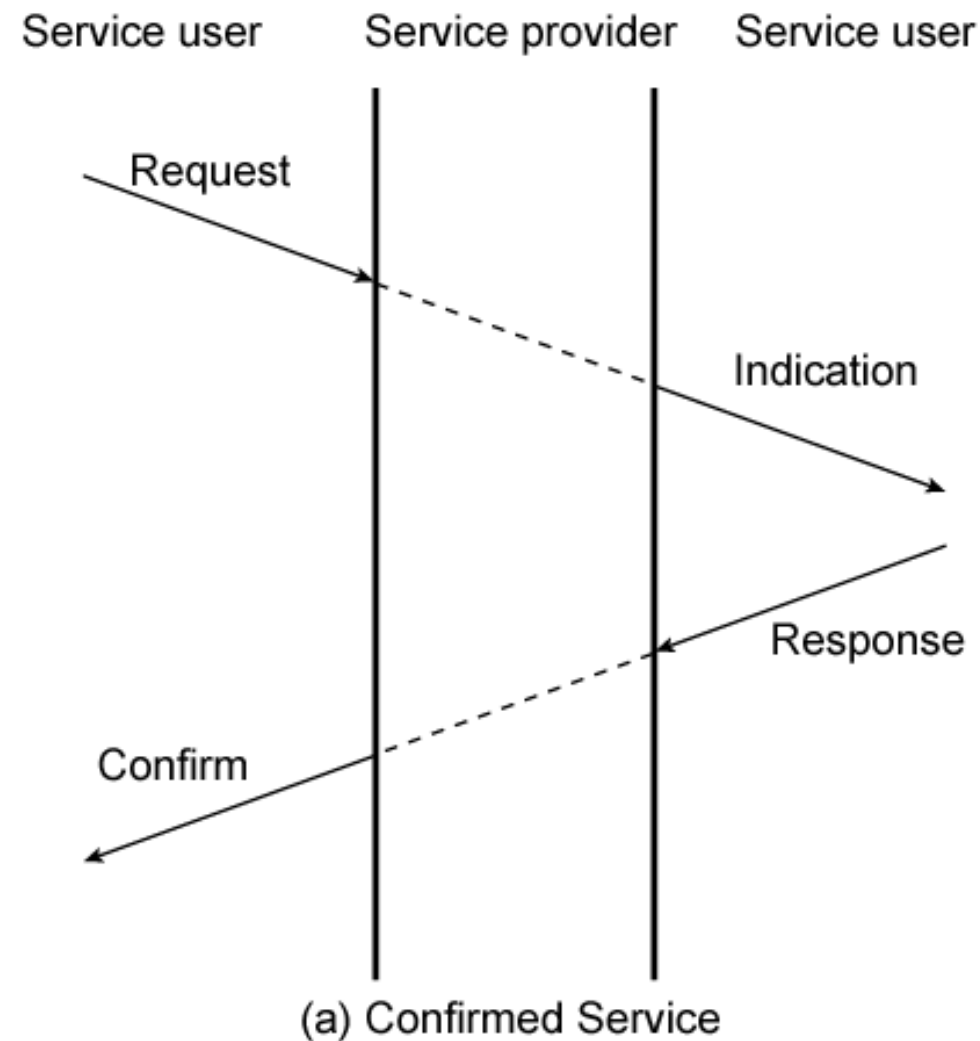
Service Primitives and Parameters

- Services between adjacent layers expressed in terms of primitives and parameters
- Primitives specify function to be performed
- Parameters pass data and control info

Primitive Types

REQUEST	A primitive issued by a service user to invoke some service and to pass the parameters needed to specify fully the requested service
INDICATION	A primitive issued by a service provider either to: indicate that a procedure has been invoked by the peer service user on the connection and to provide the associated parameters, or notify the service user of a provider-initiated action
RESPONSE	A primitive issued by a service user to acknowledge or complete some procedure previously invoked by an indication to that user
CONFIRM	A primitive issued by a service provider to acknowledge or complete some procedure previously invoked by a request by the service user

Timing Sequence for Service Primitives



OSI Layers (1)

- Physical
 - Physical interface between devices
 - Mechanical
 - Electrical
 - Functional
 - Procedural
- Data Link
 - Means of activating, maintaining and deactivating a reliable link
 - Error detection and control
 - Higher layers may assume error free transmission

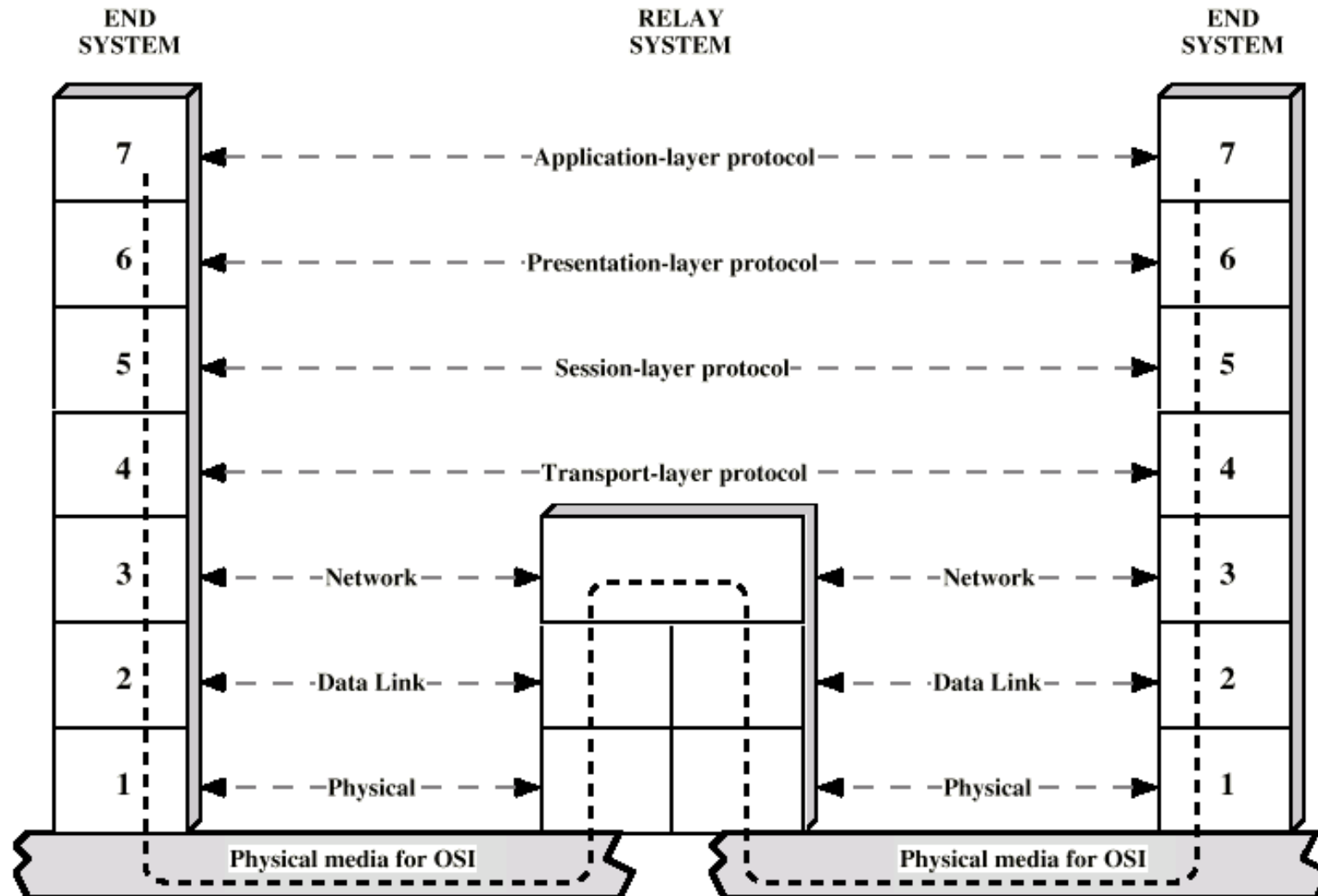
OSI Layers (2)

- Network
 - Transport of information
 - Higher layers do not need to know about underlying technology
 - Not needed on direct links
- Transport
 - Exchange of data between end systems
 - Error free
 - In sequence
 - No losses
 - No duplicates
 - Quality of service

OSI Layers (3)

- Session
 - Control of dialogues between applications
 - Dialogue discipline
 - Grouping
 - Recovery
- Presentation
 - Data formats and coding
 - Data compression
 - Encryption
- Application
 - Means for applications to access OSI environment

Use of a Relay



TCP/IP Protocol Architecture

- Developed by the US Defense Advanced Research Project Agency (DARPA) for its packet switched network (ARPANET)
- Used by the global Internet
- No official model but a working one.
 - Application layer
 - Host to host or transport layer
 - Internet layer
 - Network access layer
 - Physical layer

Physical Layer

- Physical interface between data transmission device (e.g. computer) and transmission medium or network
- Characteristics of transmission medium
- Signal levels
- Data rates
- etc.

Network Access Layer

- Exchange of data between end system and network
- Destination address provision
- Invoking services like priority

Internet Layer (IP)

- Systems may be attached to different networks
- Routing functions across multiple networks
- Implemented in end systems and routers

Transport Layer (TCP)

- Reliable delivery of data
- Ordering of delivery

Application Layer

- Support for user applications
- e.g. http, SMTP

OSI v TCP/IP

OSI	TCP/IP
Application	Application
Presentation	
Session	
Transport	Transport (host-to-host)
Network	Internet
Data Link	Network Access
Physical	Physical

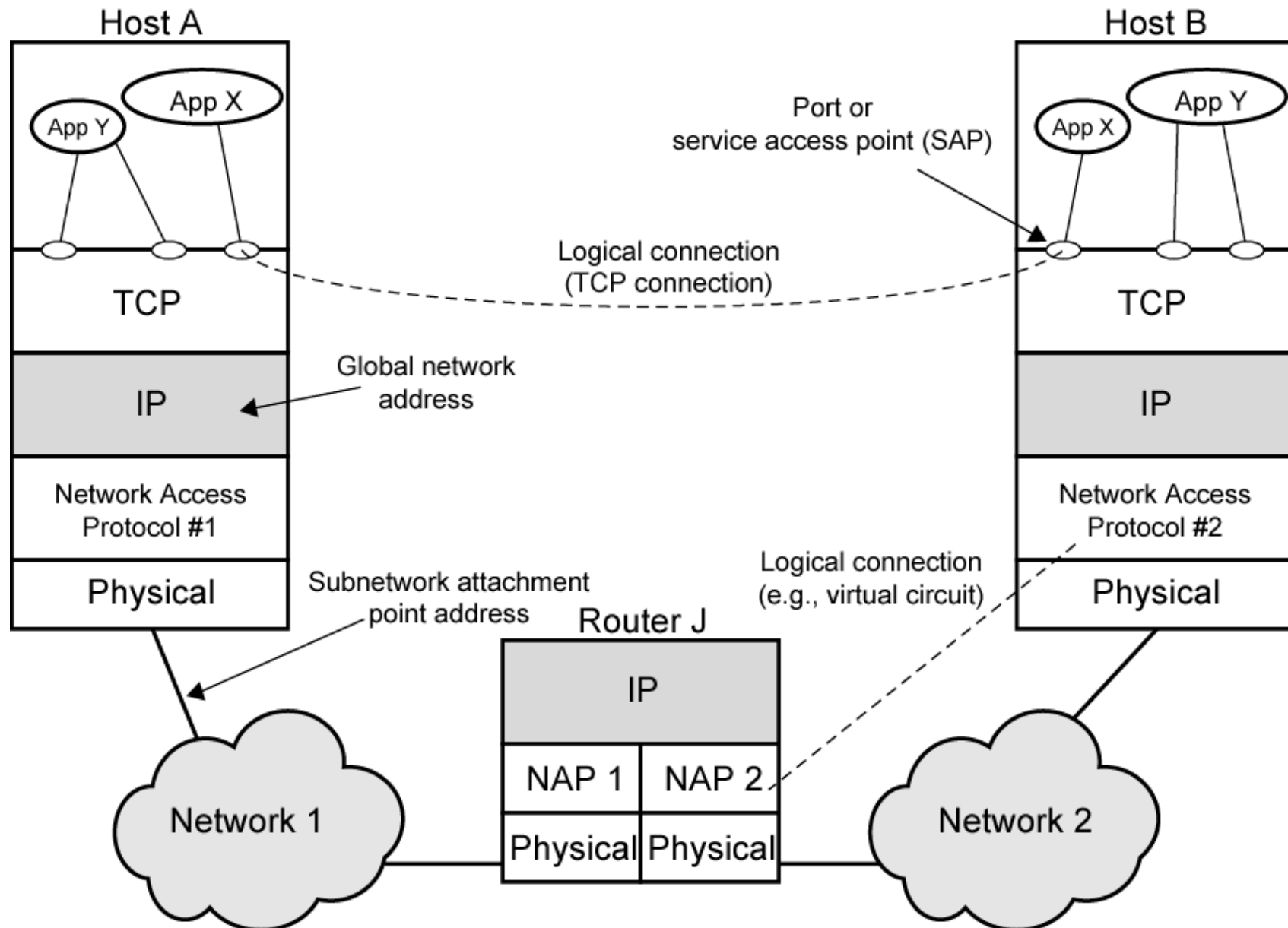
TCP

- Usual transport layer is Transmission Control Protocol
 - Reliable connection
- Connection
 - Temporary logical association between entities in different systems
- TCP PDU
 - Called TCP segment
 - Includes source and destination port (c.f. SAP)
 - Identify respective users (applications)
 - Connection refers to pair of ports
- TCP tracks segments between entities on each connection

UDP

- Alternative to TCP is User Datagram Protocol
- Not guaranteed delivery
- No preservation of sequence
- No protection against duplication
- Minimum overhead
- Adds port addressing to IP

TCP/IP Concepts



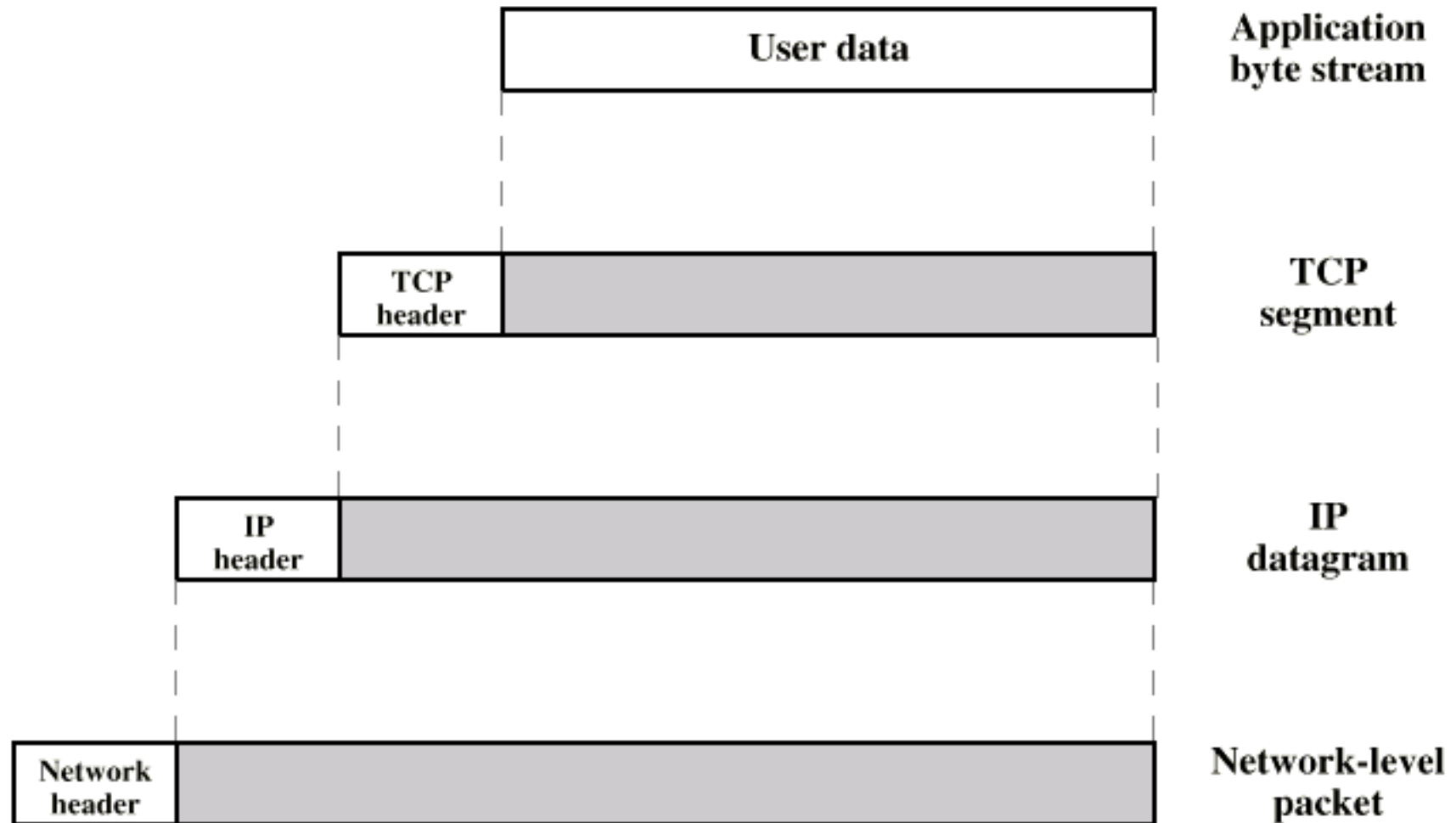
Addressing level

- Level in architecture at which entity is named
- Unique address for each end system (computer) and router
- Network level address
 - IP or internet address (TCP/IP)
 - Network service access point or NSAP (OSI)
- Process within the system
 - Port number (TCP/IP)
 - Service access point or SAP (OSI)

Trace of Simple Operation

- Process associated with port 1 in host A sends message to port 2 in host B
- Process at A hands down message to TCP to send to port 2
- TCP hands down to IP to send to host B
- IP hands down to network layer (e.g. Ethernet) to send to router J
- Generates a set of encapsulated PDUs

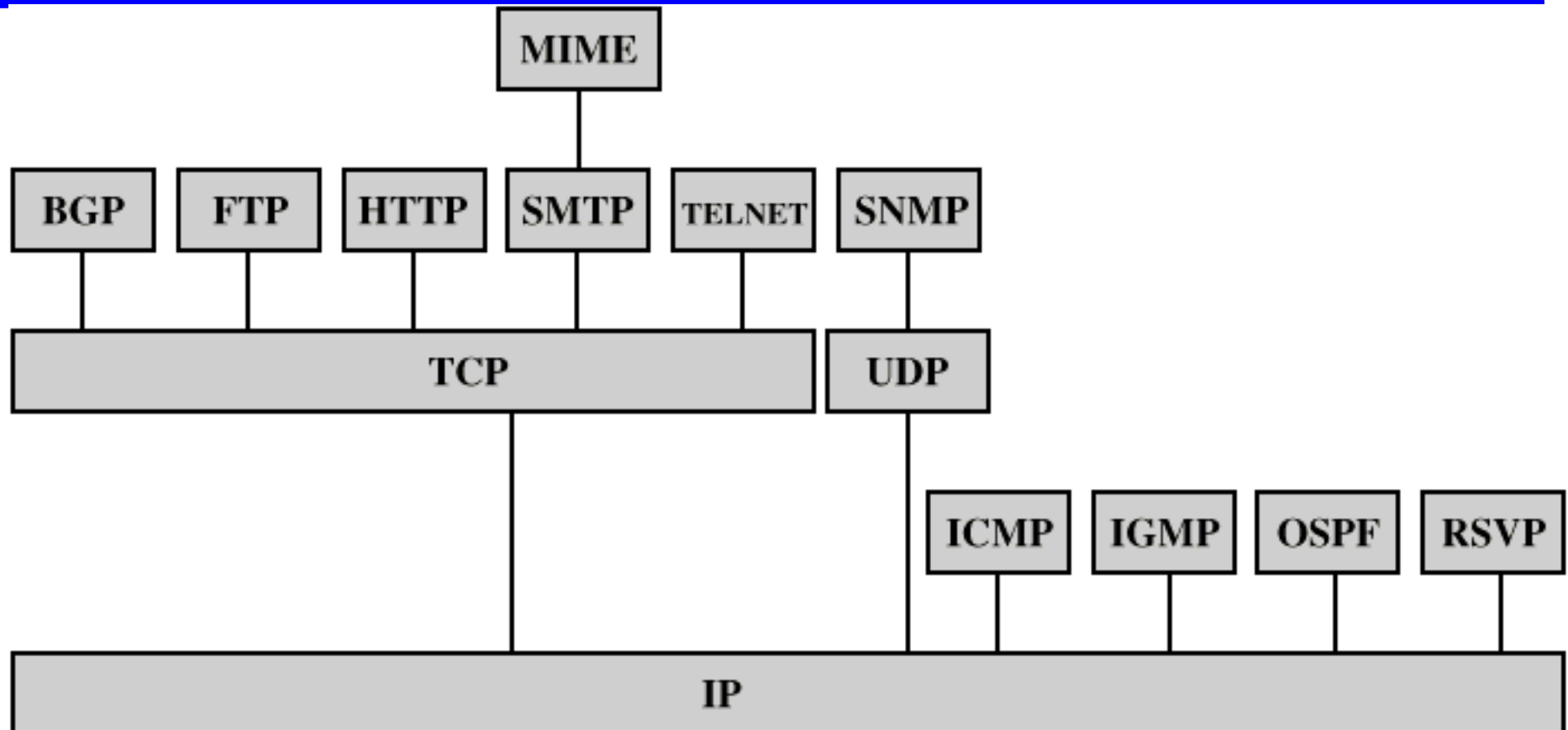
PDU in TCP/IP



Example Header Information

- Destination port
- Sequence number
- Checksum

Some Protocols in TCP/IP Suite



BGP = Border Gateway Protocol

FTP = File Transfer Protocol

HTTP = Hypertext Transfer Protocol

ICMP = Internet Control Message Protocol

IGMP = Internet Group Management Protocol

IP = Internet Protocol

MIME = Multi-Purpose Internet Mail Extension

OSPF = Open Shortest Path First

RSVP = Resource ReSerVation Protocol

SMTP = Simple Mail Transfer Protocol

SNMP = Simple Network Management Protocol

TCP = Transmission Control Protocol

UDP = User Datagram Protocol

Required Reading

- Stallings chapter 2